

BST Physics Curriculum Vol 11 Chapter 12 — Number-Theoretic Foundations Synthesis v0.4 SIGNATURE (refilled per Cal #104)

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2026-05-23 Saturday EDT (Cal #104 refill)

Vol 11 Chapter 12 — Number-Theoretic Foundations Synthesis

Chapter motivation

Vol 11 chapters 6-10 explore deep number-theoretic structures intersecting with BST substrate: - Ch 6: K3 surfaces + Niemeier lattices + Leech + Mathieu M_{24} (5-family Bridge Object architecture) - Ch 7: Heegner numbers + Cremona 49a1 elliptic curve (Stark-Heegner 1952-1967) - Ch 8: Monster moonshine + Ogg supersingular primes (Borcherds 1992 Fields Medal) - Ch 9: Cyclotomic $Q(\zeta_N)$ + Reed-Solomon (K59 substrate-coding framework) - Ch 10: Mersenne primes + Lucas-Lehmer (T2451 Sub-Substrate Mersenne Hierarchy)

This synthesis chapter unifies these threads into a coherent multi-level integer web anchoring D_{IV}^5 as substrate. The web has multiple levels: BST primary integers $\rightarrow$ Heegner discriminants $\rightarrow$ Mersenne primes $\rightarrow$ Monster supersingular primes $\rightarrow$ Stark small-primary subset — all centered on D_{IV}^5 via Cremona 49a1 + cosmological $t_{\text{cosmo}} = 47$.

Cal #99 authoritative 7-candidate enumeration (Vol 0 Ch 9): $C7 + C9 + C15 + C16 + C17a + C17b + C18$ — each candidate criterion uses NT structure from Vol 11 Ch 6-10.

Section 12.0b — Reader-grade 3-level pedagogy

Level 1: Synthesis Vol 11 Ch 6-10: K3 + Heegner + Monster + cyclotomic + Mersenne + Stark unified as multi-level integer web centered on D_{IV}^5 ; Cal #99 7-candidate enumeration cross-link ($C7 + C9 + C15 + C16 + C17a + C17b + C18$).

Level 2 (graduate-mathematician): Vol 11 chapters 6-10 explore deep number-theoretic structures intersecting with BST substrate. Multi-level integer web

centered on D_{IV}^5 : (1) BST primary integers $\{\text{rank}=2, N_c=3, n_C=5, C_2=6, g=7\} + N_{\text{max}}=137$ (Vol 0 Ch 2 + Ch 9 11 RIGOROUSLY CLOSED Strong-Uniqueness criteria); (2) Heegner discriminants $\{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ with Cremona 49a1 = BST canonical elliptic curve at conductor $g^2 = 49$, discriminant $-g^3 = -343, j = -(N_c \cdot n_C)^3 = -15^3$, CM by $Q(\sqrt{-g}) = Q(\sqrt{-7})$ — small-primary subset $\{-3, -7, -11\}$ matches BST primaries N_c, g, c_2 (Vol 11 Ch 7 + Vol 0 Ch 9 C9); (3) Mersenne primes 6/9 BST primary indices produce Mersenne primes (T2466 absorbed into Vol 0 Ch 9 C15 Sub-Substrate Mersenne Hierarchy); $g = 7 + M_g = 127$ give substrate-tick GF(128) field (Vol 11 Ch 9 + Ch 10); (4) Monster supersingular primes 15 Ogg primes; $t_{\text{cosmo}} = g^2 - \text{rank} = 47$ is one of 15 supersingular primes + factor of $196883 = 47 \cdot 59 \cdot 71$ Monster smallest non-trivial irrep dimension (Vol 4 Ch 4 T1924 + Vol 11 Ch 8); (5) Cyclotomic $Q(\zeta_{128})$ Galois group order $\varphi(128) = 64 = 2^{C_2}$ matches BST primary (Vol 11 Ch 9). Cal #99 authoritative 7-candidate Strong-Uniqueness enumeration (Vol 0 Ch 9): each candidate uses NT structure from Vol 11 — C7 Bridge Object tier (K3 + 49a1 + Q^5 central hubs); C9 Stark anchor $\{-3, -7, -11\}$; C15 Sub-Substrate Mersenne Hierarchy (absorbing T2466 BST primary Mersenne-prime density); C16 Universal α -Analog Formula (T2456 + T2462 25 HSDs); C17a TYPE I Overdetermined-Form Cluster; C17b TYPE II Cross-Domain Anchor Cluster; C18 D_{IV}^5 Rigidity Principle (T2467 + T2468). Cross-domain anchoring: integer 47 appears in (a) T1485 Λ exponent ($\Lambda = g \cdot \exp(-C_2 \cdot t_{\text{cosmo}})$); (b) M_{Pl}/m_e Bergman point; (c) H_∞ Friedmann; (d) Ogg supersingular; (e) $47 \cdot 59 \cdot 71 = 196883$ Monster — 5 cross-domain edges at single integer (T1924). Multi-level integer web is over-determined substrate signature: substrate “naturally picks” integers participating in maximal cross-domain anchoring patterns. Per Cal #99 7-candidate enumeration + Calibration #19 STANDING RULE: external register uses ratified-state count (11 RIGOROUSLY CLOSED) only; candidate-path enumeration internal/audit-chain-facing.

Level 3 (5th-grader accessible): Vol 11 Ch 6-10 explore deep number-theoretic structures (K3 surfaces, Heegner numbers, Monster moonshine, cyclotomic fields, Mersenne primes) — all interconnect with BST substrate D_{IV}^5 via a multi-level integer web. BST integers (2, 3, 5, 6, 7) connect to Heegner numbers ($\{1, 2, 3, 7, 11, \dots\}$) via Cremona 49a1 canonical elliptic curve; to Mersenne primes (6/9 BST indices give Mersenne primes); to Monster supersingular primes (cosmological scale $t_{\text{cosmo}} = 47$ is one of 15 Ogg primes); to cyclotomic $Q(\zeta_{128})$ Galois group of order $64 = 2^{C_2}$. Cal #99 lists 7 candidate Strong-Uniqueness criteria — each uses this NT web.

Section 12.1 — Vol 11 Ch 6-10 Recap

- Ch 6: K3 + Niemeier + Leech + Mathieu M_{24} (5-family Bridge Object Vol 0 Ch 9 C11)
- Ch 7: Heegner + Cremona 49a1 (Stark Vol 0 Ch 9 C9)
- Ch 8: Monster + Ogg supersingular (Vol 4 Ch 4 T1924)
- Ch 9: Cyclotomic $Q(\zeta_{128})$ + Reed-Solomon (K59 + Vol 1 Ch 10)
- Ch 10: Mersenne primes + Lucas-Lehmer (T2451 + T2466 Vol 0 Ch 9 C15)

Section 12.2 — Multi-Level Integer Web Centered on D_{IV}^5

Level 1: BST primary integers $\{\text{rank}=2, N_c=3, n_C=5, C_2=6, g=7\} + N_{\text{max}}=137$. Level 2: Heegner discriminants anchoring Cremona 49a1 = $(g^2, -g^3, -(N_c \cdot n_C)^3, Q(\sqrt{-g}))$. Level 3: Mersenne primes 6/9 BST indices Mersenne-rich; $M_g = 127$ substrate-tick GF(128). Level 4: Monster supersingular $t_{\text{cosmo}} = 47 + 196883 = 47 \cdot 59 \cdot 71$. Level 5: Cyclotomic $Q(\zeta_{128})$ Galois order $64 = 2^{C_2}$.

Section 12.3 — Cal #99 7-Candidate Enumeration Cross-Link

Vol 0 Ch 9 Strong-Uniqueness Theorem Cal #99 authoritative 7-candidate enumeration: - C7 Bridge Object tier (Vol 11 Ch 6-7 K3 + 49a1 + Q^5 hubs) - C9 Stark anchor $\{-3, -7, -11\}$ (Vol 11 Ch 7) - C15 Sub-Substrate Mersenne Hierarchy (Vol 11 Ch 10; absorbs T2466) - C16 Universal α -Analog Formula 25 HSDs - C17a TYPE I Overdetermined-Form Cluster (Task #244 cluster TYPES) - C17b TYPE II Cross-Domain Anchor Cluster (Task #244) - C18 D_{IV}^5 Rigidity Principle (T2467 + T2468 Friday)

Section 12.4 — Cross-Domain Anchoring at $t_{\text{cosmo}} = 47$

Per Vol 4 Ch 4 T1924: integer 47 appears in 5 independent cross-domain edges: - (a) T1485 Λ exponent ($\Lambda = g \cdot \exp(-C_2 \cdot t_{\text{cosmo}})$ at 0.076 dex) - (b) M_{Pl}/m_e Bergman point at $45 = t_{\text{cosmo}} - \text{rank}$ - (c) H_∞ Friedmann - (d) Ogg supersingular - (e) $47 \cdot 59 \cdot 71 = 196883$ Monster smallest non-trivial irrep

Section 12.5 — Over-Determined Substrate Signature

Multi-level integer web is over-determined: substrate “picks” integers participating in maximal cross-domain anchoring patterns.

Substrate-cartography reading per Vol 11 Ch 11: D_{IV}^5 is generative geometric object; multi-level integer web is structural signature.

Section 12.6 — Cal #99 + Cal #19 Register Discipline

Per Calibration #19 STANDING RULE: external register uses ratified-state count (11 RIGOROUSLY CLOSED) only.

Candidate-path enumeration (7 candidates per Cal #99) is internal/audit-chain-facing.

Section 12.7 — Honest scope + Connection

- Vol 11 Ch 6-10 synthesis [\[?\]](#)
- Multi-level integer web centered on D_{IV}^5 [\[?\]](#)
- Cal #99 7-candidate enumeration cross-link [\[?\]](#)

- T1924 5 cross-domain edges at $t_{\text{cosmo}} = 47$

Connection: - Vol 11 Ch 6-10 (all NT chapters) - Vol 0 Ch 9 Strong-Uniqueness Theorem - Vol 4 Ch 4 T1924 + T1485 cosmological anchor - Vol 11 Ch 11 Generative Geometry Framing

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